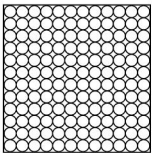
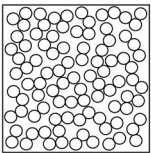
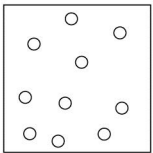


4.1.1 States of matter

The model of particles in motion can be used to account for states of matter, differences in density, the pressure of gases, and changes of state. This model is applied in [Transport into and out of cells](#) (page 30) to explain how substances are transported into and out of cells through diffusion and osmosis, and in [Systems in the human body](#) (page 36), where it is applied to substances crossing exchange surfaces. The nature of the particles (atoms, molecules and ions) is examined in more detail in [Atomic structure](#) (page 24) and [Structure and bonding](#) (page 98).

There are two required practicals: one to study the density of solid and liquid objects, another to investigate energy transfers by measuring the specific heat capacity of materials.

4.1.1.1 A particle model

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall and explain the main features of the particle model in terms of the states of matter and change of state, distinguishing between physical and chemical changes. (HT only) Explain the limitations of the particle model in relation to changes of state when particles are represented by inelastic spheres.	The three states of matter are solid, liquid and gas. Melting and freezing take place at the melting point, boiling and condensing take place at the boiling point. The three states of matter can be represented by a simple model. In this model, particles are represented by small solid spheres. Particle theory can help to explain melting, boiling, freezing and condensing.    Solid Liquid Gas (HT only) Limitations of the simple model include that there are no forces between the spheres, and that atoms, molecules and ions are not solid spheres.	WS 1.2 Recognise/draw simple diagrams to model the difference between substances in the solid, liquid and gas states. WS 3.5 Predict the states of substances at different temperatures given appropriate data. MS 1d Relate the size and scale of atoms to objects in the physical world.

4.1.1.2 Density

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Define density and explain the differences in density between the different states of matter in terms of the arrangements of the atoms or molecules.	The density of a material is defined by the equation: $density = \frac{mass}{volume}$ $\left[\rho = \frac{m}{V} \right]$ density, ρ, in kilograms per metre cubed, kg/m³ mass, m, in kilograms, kg volume, V, in metres cubed, m³	MS 1a, 1b, 1c, 3c Recall and apply this equation to changes where mass is conserved. WS 3.3 Carry out and represent mathematical and statistical analysis. WS 4.3, 4.5 Use and interconvert SI units in calculations.

Required practical activity 1: use appropriate apparatus to make and record the densities of regular and irregular solid objects and liquids. Volume should be determined from the dimensions of a regularly shaped object and by a displacement technique for irregularly shaped objects. Dimensions to be measured using appropriate apparatus such as a ruler, micrometer or Vernier callipers.

AT skills covered by this practical activity: physics AT 1.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 162).

4.1.1.3 Gas pressure

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how the motion of the molecules in a gas is related both to its temperature and its pressure: hence explain the relation between the temperature of a gas and its pressure at constant volume (qualitative only).	The molecules of a gas are in constant random motion. The temperature of the gas is related to the average kinetic energy of the molecules. The higher the temperature, the greater the average kinetic energy and so the faster the average speed of the molecules. When the molecules collide with the wall of their container they exert a force on the wall. The total force exerted by all of the molecules inside the container on a unit area of the walls is the gas pressure. Changing the temperature of a gas, held at constant volume, changes the pressure exerted by the gas.	

4.1.1.4 Heating and changes of state

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe how heating a system will change the energy stored within the system and raise its temperature or produce changes of state. Describe how, when substances melt, freeze, evaporate, condense or sublime, mass is conserved but that these physical changes differ from chemical changes because the material recovers its original properties if the change is reversed.	Energy is stored inside a system by the particles (atoms and molecules) that make up the system. This is called internal energy. The amount of energy needed to change state from solid to liquid and from liquid to gas depends on the strength of the forces between the particles of the substance. The nature of the particles involved depends on the type of bonding and the structure of the substance. The stronger the forces between the particles the higher the melting point and boiling point of the substance.	This topic links with Structure and bonding. (page 98) WS 3.3 Carry out and represent mathematical and statistical analysis.

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Define the term specific heat capacity and distinguish between it and the term specific latent heat.	The increase in temperature of a system depends on the mass of the substance heated, the type of material and the energy input. The following equation, given on the Physics equations sheet, applies: $\text{change in thermal energy} = \text{mass} \times \text{specific heat capacity} \times \text{temperature change}$ $[\Delta E = m c \Delta \theta]$ change in thermal energy, ΔE, in joules, J mass, m, in kilograms, kg specific heat capacity, c, in joules per kilogram per degree Celsius, J/kg °C temperature change, $\Delta \theta$, in degrees Celsius, °C The specific heat capacity of a substance is the amount of energy required to raise the temperature of one kilogram of the substance by one degree Celsius. The energy needed for a substance to change state is called latent heat. When a change of state occurs, the energy supplied changes the energy stored (internal energy) but not the temperature.	WS 3.5, MS 4a Interpret heating and cooling graphs that include changes of state. WS 4.3, 4.5, MS 1a, 3c, 3d Apply this equation, which is given on the Physics equations sheet, to calculate energy changes when a material is heated or cooled. WS 3.3 Carry out and represent mathematical and statistical analysis.
Describe and calculate the changes in energy involved when a system is changed by heating (in terms of temperature change and specific heat capacity).	The specific latent heat of a substance is the amount of energy required to change the state of one kilogram of the substance with no change in temperature. The following equation, given on the Physics equations sheet, applies: $\text{energy for a change of state} = \text{mass} \times \text{specific latent heat}$ $[E = m L]$ energy, E, in joules, J mass, m, in kilograms, kg specific latent heat, L, in joules per kilogram, J/kg Specific latent heat of fusion – change of state from solid to liquid. Specific latent heat of vaporisation – change of state from liquid to vapour.	WS 4.3, 4.5, MS 1a, 3c, 3d Apply this equation, which is given on the Physics equations sheet, to calculate energy changes during changes of state.

Required practical activity 2: an investigation to determine the specific heat capacity of one or more materials. The investigation will involve linking the decrease of one energy store (or work done) to the increase in temperature and subsequent increase in thermal energy stored.

AT skills covered by this practical activity: physics AT 1 and 5.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 163).

4.1.1.5 Meanings of purity

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain what is meant by the purity of a substance, distinguishing between the scientific and everyday use of the term 'pure'.	In chemistry, a pure substance is a single element or compound, not mixed with any other substance. Pure elements and compounds melt and boil at specific temperatures. Melting point and boiling point data can be used to distinguish pure substances from mixtures. In everyday language, a pure substance can mean a substance that has had nothing added to it, so it is unadulterated and in its natural state.	WS 3.5 Use melting point data to distinguish pure from impure substances.